

CONSTRUCT EDGE 2024



BUILDING ENGINEERING & CONSTRUCTION MANAGEMENT
RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
RAJSHAHI- 6204, BANGLADESH

আমাদের বৈশিষ্ট্য সমূহ

আমাদের রয়েছে -

- ✦ মনোমুগ্ধকর স্থায়ী ক্যাম্পাস।
- ✦ যোগ্য ও অভিজ্ঞ শিক্ষকমণ্ডলী।
- ✦ নিজস্ব পরিবহণ ব্যবস্থা।
- ✦ শিক্ষার্থীদের জন্য রয়েছে আবাসিক হাল এর সুবিধা।
- ✦ দরিদ্র ও মেধাবী শিক্ষার্থীদের জন্য ১০% থেকে ১০০% ছাড়ের ব্যবস্থা।
- ✦ শিক্ষার্থীদের মানবিক গুণাবলী বিকশিত করার জন্য রয়েছে বিভিন্ন ক্লাব ও রোভার স্কাউট গ্রুপ।



প্রোগ্রাম সমূহ

অনার্স প্রোগ্রাম

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- ইম্নামিক স্টাডিজ
- আইন
- ব্যবসা প্রশাসন (বিবিএ)
- ম্যানেজমেন্ট এন্ড ইনফর্মেশন সিস্টেমস

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- আইন
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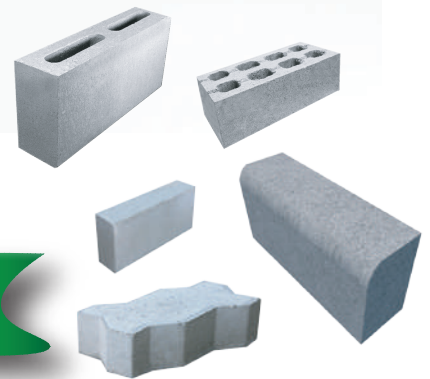
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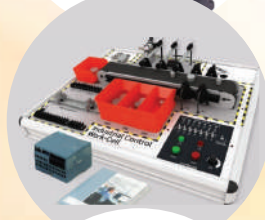
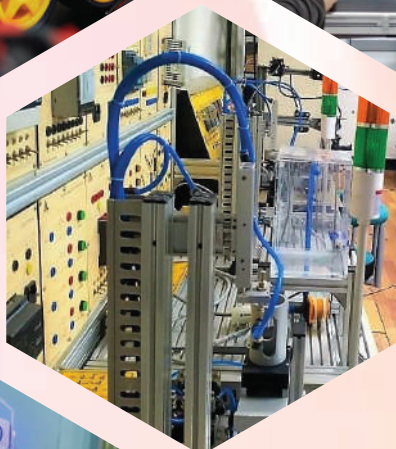
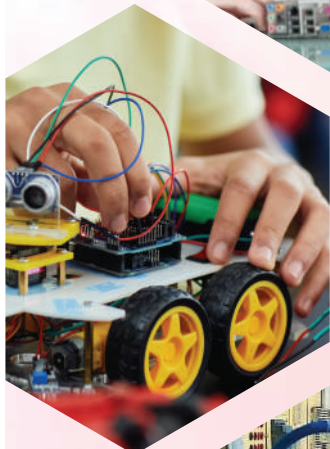
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Message...



From the Vice-Chancellor

Prof. Dr. Engr. Md. Jahangir Alam
Rajshahi University of Engineering & Technology

I am delighted to know that the Department of **Building Engineering and Construction Management (BECM)** is going to organize their first ever Construct Carnival 1.0 at Rajshahi University of Engineering & Technology (RUET) during 30th-31st May, 2024. This Carnival is going to be a celebration of innovation, collaboration, leadership and the boundless potential of building engineering and construction management.

The building engineering and construction management field of study prepares students to manage design, construction and operation of sustainable buildings and infrastructure systems in the increasingly global construction industry. Since the inception of BECM in 2016 with just 30 students, it has grown into a competent department under the Faculty of Civil Engineering. The main goal of the department has always been to help students explore different career paths and prepare them to be successful engineers and responsible citizens.

The **Construct Carnival 1.0** will be a memorable event where students can show their skills on an international level. I warmly welcome all participants and believe that your time here will be life-changing. This event will not only improve your professional skills but also help you become innovative leaders who can make a big impact on society.

I hope the **Construct Carnival 1.0** organized by the BECM will provide a platform for the students across the country to share their views and knowledge in the rapidly emerging areas of engineering and technologies using AI & Robotics and I believe that this event will also play a proactive role to improve technical knowledge, latest skill sets as well as interpersonal relationships of the students. I extend my best congratulations to all the student, teachers, staff, student coordinators and participants and wish the **Construct Carnival 1.0** a grand success.



Message...



From the Dean

Prof. Dr. Md. Kumruzzaman
Rajshahi University of Engineering & Technology

As the Dean of the Civil Faculty at Rajshahi University of Engineering & Technology (RUET), I am thrilled to announce the upcoming 'Construct Carnival 1.0', scheduled to take place on May 30th and 31st, 2024. This event marks a significant celebration of ingenuity, innovation, and unity within our academic community, and I extend a warm invitation to guests, scholars, and students alike to join us in this festive occasion. Countless hours of meticulous planning, coordination, and dedication have been invested in preparing for the Construct Carnival. It is with great confidence that I anticipate a remarkable showcase of the talent, creativity, and collaborative spirit that represent our esteemed department. As we eagerly await the arrival of the Construct Carnival, I urge all participants to embrace the ethos of unity and companionship that lies at the core of our academic community. Together, let us foster an environment of mutual support and encouragement as we come together to celebrate our collective achievements.

I extend my heartfelt gratitude to all those who have contributed to the planning and organization of the Construct Carnival, including our valued sponsors, enthusiastic participants, and dedicated attendees. It is through your collective efforts that this event has been made possible. With sincere hopes for the success of 'Construct Carnival 1.0', I eagerly anticipate the festivities and the exchange of new perspectives and ideas that will undoubtedly enrich our academic community. Let us come together to celebrate excellence and forge lasting connections that will propel us towards a brighter future.

Message...



From the Head

Prof. Dr. Md. Robiul Awall

Rajshahi University of Engineering & Technology

We are excited to announce the upcoming 'Construct Carnival 1.0', hosted by the Department of Building Engineering & Construction Management (BECM) at Rajshahi University of Engineering & Technology (RUET) on May 30th and 31st, 2024. This event is very special, celebrating creativity, innovation, and community spirit across the country. We warmly invite guests, scholars, and students to join us at this carnival.

Many hours of careful planning, coordination, and hard work have gone into preparing for this event. We are confident that it will be a great showcase of the talent, creativity, and teamwork that define our department. As we look forward to the Construct Carnival, we encourage all participants to embrace the spirit of unity and friendship that is the heart of our community.

We deeply appreciate everyone who has helped plan and organize the Construct Carnival, including our sponsors, participants, and attendees. Your efforts have made this event possible. I am deeply honored to express my heartfelt gratitude to our distinguished chief guest, Prof. Dr. Engr. Md. Jahangir Alam, Vice-Chancellor of RUET, and our esteemed special guests: Md. Amirul Haque Bhuiya, DG of BWDB, Md. Ashraful Alam, DG of HBRI, Dr. Md. Shafiul Islam, Senior Research Engineer at HBRI, and Pintu Kanungoe, Director of the River Research Institute, Faridpur. Your presence and insights have greatly enriched Construct Carnival 1.0. I also thank the students from various universities across Bangladesh and our RUET students for their enthusiastic participation. Your contributions have made this event a grand success.

With best wishes for the success of 'Construct Carnival 1.0', we look forward to the fun festivities and the new connections and ideas that will come from this celebration of creativity and excellence.

Message...

From the Director General
Md Ashraful Alam
Housing and Building Research Institute



Good morning to you all. It is with great pleasure that I stand before you today at the inaugural 'Construct Carnival 1.0' at Rajshahi University of Engineering & Technology (RUET). I am deeply honored to be invited as a special guest, and I extend my heartfelt thanks to the RUET BECM for this opportunity. 'Construct Carnival 1.0' is a momentous occasion that celebrates the talents and ingenuity of students in the fields of Civil Engineering. It is an event that fosters creativity, collaboration, and innovation, providing a platform for students to showcase their skills and ideas. Today, we witness an array of contests designed to challenge and inspire participants. These contests not only test technical prowess but also encourage critical thinking, problem-solving, and effective communication.

As the Director General of the Housing and Building Research Institute (HBRI), I understand the importance of nurturing talent and fostering excellence in the construction and engineering sectors. Events like 'Construct Carnival 1.0' play a crucial role in preparing students for the challenges and opportunities that lie ahead in their careers. They provide practical experience, encourage innovation, and promote collaboration – all essential qualities for success in today's world. I commend the RUET BECM Club, the faculty, and all the participants for their hard work and dedication in organizing this event. Your efforts have created a vibrant and inspiring atmosphere for learning and growth. To the students participating in this carnival, I encourage you to seize this opportunity to learn, grow, and connect with your peers. Embrace the challenges, push your boundaries, and strive for excellence. The skills and experiences you gain here will undoubtedly shape your future and contribute to the advancement of our society.

I wish 'Construct Carnival 1.0' great success. May it serve as a catalyst for innovation and positive change in the field of construction and Engineering.

A portrait of Md. Amirul Haque Bhuiya, a middle-aged man with grey hair, wearing a dark blue suit, white shirt, and a red and white striped tie. The portrait is set within a hexagonal frame with an orange border. In the background, there is a stylized orange sun and some geometric line art.

Message...

From the Director General

Md. Amirul Haque Bhuiya

Bangladesh Water Development Board

I am honored to be here today at the first 'Construct Carnival 1.0' at the department of BECM RUET. Thank you to RUET BECM for inviting me to this exciting event. 'Construct Carnival 1.0' is a wonderful platform that brings together bright minds from across the country. This event celebrates innovation, creativity, and teamwork in fields like Civil Engineering, Building Engineering & Construction Management, Urban and Regional Planning, and Architecture. Today's event includes many exciting contests. As the Director General of the Bangladesh Water Development Board, I know how important skilled engineers and planners are for our country's growth. Events like this help prepare students for the real-world challenges they will face in their careers. They provide practical experience and the chance to apply what they have learned in class.

Besides this, I want to thank the RUET BECM Club, the faculty, and all the participants for their hard work in making this event possible. Your dedication has created a great environment for learning and innovation. To all the students, make the most of this opportunity. Take part in the workshops and activities, learn from each other, and challenge yourselves. The skills and knowledge you gain here will be very valuable in your future careers. In closing, I wish 'Construct Carnival 1.0' great success. Let's continue to work together to build a better future.

Message...

From the Director General

Pintu Kanungoe
River Research Institute, Faridpur



Greetings to all. I am truly delighted to be here today as a special guest at the inaugural 'Construct Carnival 1.0' at Rajshahi University of Engineering & Technology (RUET). I extend my heartfelt gratitude to the RUET BECM for this gracious invitation. 'Construct Carnival 1.0' is a wonderful opportunity to celebrate the creativity and talent of students in the fields of Civil Engineering arena. It is a platform that encourages innovation, collaboration, and learning, and I am thrilled to be a part of it. Today, we witness a series of contests designed to challenge and inspire participants. These contests showcase the diverse skills and expertise of our future engineers and planners.

As the Director of the River Research Institute in Faridpur, I understand the importance of research and innovation in addressing the challenges facing our water resources. Events like 'Construct Carnival 1.0' play a crucial role in nurturing talent and fostering excellence in the field of construction and engineering. They provide students with valuable hands-on experience and encourage them to think critically and creatively. I commend the RUET BECM, the faculty, and all the participants for their dedication and hard work in organizing this event. Your efforts have created a vibrant and inspiring atmosphere for learning and growth. To the students participating in this carnival, I encourage you to make the most of this opportunity. Embrace the challenges, collaborate with your peers, and push the boundaries of your knowledge and skills. The experiences you gain here will undoubtedly shape your future and contribute to the advancement of our society.

A portrait of Mehedi Hasan, a man with glasses and a checkered shirt, is framed in a hexagonal shape with a thick orange border. The background of the portrait is a purple and pink gradient. Above the portrait is a solid orange circle. The entire graphic is set against a background of thin, intersecting black lines.

Message...

From the Advisor

Mehedi Hasan
BECM Club RUET

Ladies and gentlemen, esteemed guests, faculty members, and dear students, It is my pleasure to welcome you all to the first 'Construct Carnival 1.0' at the dept. of BECM, RUET. As an advisor of the BECM Club and a faculty member of the department of BECM, I am thrilled to see this event come to life. 'Construct Carnival 1.0' is a special occasion that brings together students from various disciplines to celebrate innovation, creativity, and collaboration. This event showcases the talent and hard work of our students. It is a platform where students can demonstrate their skills, share their ideas, and learn from each other.

Today, we have several exciting contests lined up: Cad Expert, Mecha Mind, Poster Presentation, Archi capture, Render Rampage, and Smart Management Maestro. These contests are designed to challenge and inspire you, helping you grow both personally and professionally. I want to take this opportunity to praise the RUET BECM Club, the faculty, and all the participants for their commitment and effort in arranging this event. Your dedication has made 'Construct Carnival 1.0' possible and created an environment that encourages learning and innovation. To the students, I encourage you to make the most of this opportunity. Participate actively, collaborate with your peers, and push the boundaries of your knowledge and skills. The experiences you gain here will be invaluable in your future careers and will help you contribute to the advancement of our society.

Let 'Construct Carnival 1.0' be a smashing success. I hope it motivates you to keep pushing yourself to your limits and to make a positive impact on the world.

Message...

From the President

Aditty Arshad
BECM Club RUET



As the President of the RUET BECM Club, it is an honor to announce our exciting new event, 'Construction Carnival 1.0'. This nationwide event will spark passions, build connections, and create unforgettable moments for students in Civil Engineering, Building Engineering & Construction Management, Urban and Regional Planning, and Architecture. It is a great chance to showcase excellence, innovation, and research in building a better world for everyone.

I want to thank each and every participant who has given their time and expertise to make this vision a reality. Your contributions are key to the success of this program. Even though this is our first nationwide event, the dedication and enthusiasm of our club members have made it possible to organize this program perfectly.

With engaging workshops, insightful discussions, academic activities, and more, there is something for everyone to enjoy. May 'Construction Carnival 1.0' be a memorable and enriching experience for all of us and serve as a source of inspiration for future engineers.



About BECM

In 2016, the Rajshahi University of Engineering & Technology (RUET) established the Building Engineering & Construction Management (BECM) department, a bold response to the critical challenges plaguing Bangladesh's construction industry. Starting with a dedicated cohort of 30 students, BECM has evolved into a dynamic hub for education, research, and innovation, dedicated to advancing the technical performance of buildings and construction management practices.

Bangladesh's rapid development and high population density have exacerbated issues such as inadequate construction safety, inefficient management, and a lack of eco-friendly infrastructure. BECM addresses these head-on, aiming to produce graduates with the comprehensive expertise needed to tackle these challenges and ensure a sustainable future for the nation's-built environment.

The BECM curriculum goes far beyond traditional engineering, encompassing a diverse range of topics crucial for building professionals. Students delve into project management, structural and foundation engineering, construction safety, earthquake engineering, architecture, digital building management, resource efficiency, and HVAC control. This holistic approach equips graduates to lead complex construction projects and make significant contributions to the industry, armed with a robust foundation in construction technology, architecture, and structural design.



CONSTRUCT
CARNIVAL 1.0

About Construct Carnival

Construct Carnival is a prestigious national competition organized by the Department of Building Engineering and Construction Management (BECM) at Rajshahi University of Engineering and Technology (RUET). It serves as a platform for the students from all over the country to showcase their talents, skills, and innovative ideas in the realm of Civil Engineering, Architecture, Urban and Regional Planning (URP), and Construction Management. The primary objective of Construct Carnival is to foster creativity, innovation, and excellence in the field of civil engineering and related disciplines. By bringing together students, academics, and industry professionals, the event aims to promote collaboration, knowledge sharing, and hands-on learning experiences. By providing a platform for students to demonstrate their skills and learn from industry experts, this national event plays a crucial role in shaping the future of the built environment. Participants get the chance to interact with peers from other institutions, as well as professionals from the industry. This exalted festival features the following key segments:

Cad Expert
Render Rampage
Mechamind
Idea Contest
Poster presentation
Smart Management Maestro
Archi capture



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Self-Sensing Concrete: Revolutionizing Civil Infrastructure

Md. Mustakim
Dept. of BECM, RUET

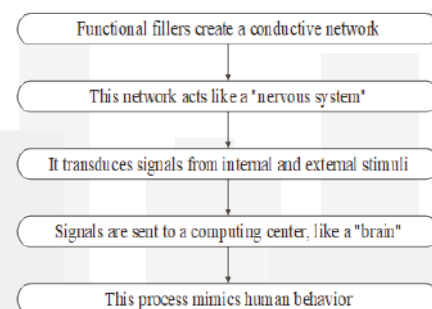
Self-Sensing Concrete (SSC) is a structural material that inherently monitors its own condition, eliminating the need for external sensors. This groundbreaking concrete can sense internal and environmental variations such as stress, strain, cracks, damage, temperature, and humidity by integrating functional fillers or sensing components. SSC is classified into two main types: intrinsic self-sensing concrete and non-intrinsic self-sensing concrete. Intrinsic self-sensing concrete integrates conductive materials like carbon fibers into the concrete mix, enabling it to monitor its own structural health. It detects stress, strain, cracks, temperature, and humidity changes through variations in electrical resistance, providing real-time data for proactive maintenance and enhanced safety without external sensors. On-intrinsic self-sensing concrete uses external sensors, such as embedded fiber optics or piezoelectric devices, to monitor structural health. These sensors detect stress, strain, cracks, and environmental changes, transmitting data for analysis. This method requires additional installation and maintenance but effectively enhances structural monitoring and safety.

Composition of Self-Sensing Concrete

The materials used to fabricate SSC are varied, making the selection of suitable materials and their proportions crucial. SSC, as a composite material, primarily includes matrix materials like conventional concrete and functional fillers. Additionally, auxiliary materials might be necessary to properly disperse these functional fillers within the matrix. The matrix material, which can be concrete, mortar, or paste, binds the functional fillers, forming the bulk of the composite. The prevalent utilization of Portland cement as a binder makes it the usual choice for the matrix material in SSC. The functional filler is a vital component of SSC, as it determines the sensing properties of the composite.

Principles of Self Sensing Concrete

Sensing concrete operates on the principle of a conductive network, akin to a "nervous system," composed of functional fillers. This network transduces signals from internal and external stimuli to a central computing unit, resembling a "brain," mirroring human behavior. Unlike traditional sensors, sensing concrete thrives in challenging environments, prolonging infrastructure lifespan. Its integration into concrete structures enables smart infrastructure, enhancing safety, reliability, and durability.



Sensing Mechanism

The self-sensing capability of concrete is primarily based on its electrical resistivity. By incorporating conductive materials, the electrical resistivity of the concrete changes in response to mechanical strain. These changes can be monitored to infer the stress and damage levels within the structure. There are two primary mechanisms through which self-sensing concrete operates: Piezo resistivity refers to a material's resistivity that varies with strain, converting external loads into changes in electrical resistivity. Widely utilized in self-sensing concrete, it offers superior accuracy compared to other techniques. By incorporating conducting fillers into the matrix, both strength and sensing parameters are enhanced. The change in resistivity results from dimensional alterations under strain or stress. Also, Self-sensing concrete exhibits electromechanical response where strain causes a measurable change in electrical resistance due to the deformation of its conductive network. Cracks disrupt this network, increasing resistance and allowing early damage detection. Additionally, stress changes affect electrical properties, enabling real-time monitoring of structural health.

Conductive Mechanism

Producing SSC entails incorporating functional fillers into a concrete matrix, with the conductivity of the material closely linked to the filler concentration. The conductive characteristic curve, which shows the relationship between SSC's electrical resistivity and filler concentration, has three distinct zones. Zone A, the "insulation zone," exhibits high resistivity. In this zone, the filler

concentration is far below the percolation threshold, leading to large gaps between fillers and minimal aggregation, making it difficult to form conductive paths. Zone B, the "percolation zone," is characterized by a rapid decrease in resistivity as the filler concentration increases, enabling the formation of conductive paths and significantly enhancing conductivity. Zone C, the "conductive zone," occurs when the filler concentration is well above the percolation threshold, resulting in nearly total contact between fillers and maintaining low resistivity.

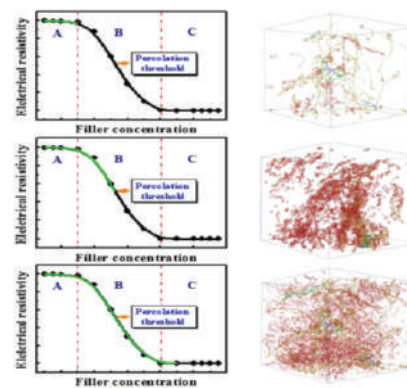


Fig 01: Change of electrical resistivity

Applications of Self Sensing Concrete

Monitoring the health of structures is essential to uphold the safety, longevity, efficiency, and environmental responsibility of civil infrastructure. Traditional sensors encounter challenges related to durability and suitability for concrete settings. Self-sensing concrete (SSC) resolves these issues by embedding sensor capabilities directly into the structural material. SSC employs electrical resistance measurements for continuous monitoring of structural condition, detecting indicators like stress, strain, cracks, and damage in real time. Different arrangements, such as bulk, coating, sandwich, bonded, and embedded configurations, optimize monitoring effectiveness while minimizing expenses. In traffic management, SSC-based vehicle detection sensors offer easy installation, large detection areas, cost-effectiveness, and robustness. These sensors enhance structural health monitoring and condition assessment, supporting sustainable infrastructure and efficient traffic management.

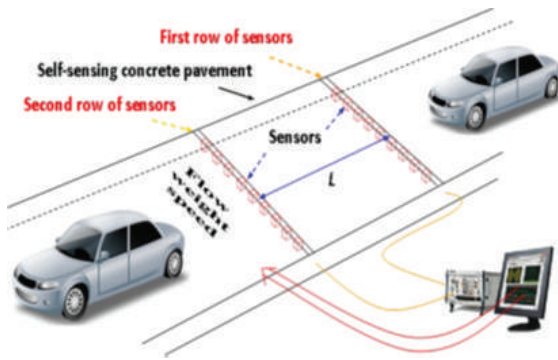


Fig 02: SSC pavement structure for vehicle detection

Advantages & Challenges of Self Sensing Concrete

Self-sensing concrete (SSC) offers significant advantages in infrastructure monitoring and maintenance. One of its primary benefits is its ability to detect and monitor structural integrity in real-time, reducing the need for external sensors and enabling proactive maintenance. This leads to enhanced safety and longevity of structures such as bridges, buildings, and pavements. SSC's inherent ability to detect strain, stress, cracks, and temperature changes facilitates early detection of potential issues, thus preventing catastrophic failures. Additionally, the integration of SSC can lead to cost savings by reducing the frequency and complexity of inspections, as well as lowering the overall lifecycle costs of structures. The development and use of SSC present several hurdles. Ensuring the consistent dispersion of functional fillers in the concrete matrix is crucial for dependable sensing. The elevated cost of these materials compared to traditional concrete could hinder widespread acceptance. Long-term durability and stability under diverse environmental conditions are still under investigation. Technical obstacles include data calibration and interpretation, necessitating specialized expertise and possibly new monitoring standards. Nonetheless, SSC holds promise for advancing smart infrastructure.

In conclusion, Self-Sensing Concrete (SSC) stands as a transformative technology with the potential to revolutionize civil infrastructure. With its ability to detect stress, damage, and structural issues in real time, SSC enhances safety, sustainability, and efficiency in construction projects. As research and development efforts continue, SSC holds promise for creating a more resilient and sustainable future for civil infrastructure worldwide.

Overcoming Power Challenges in Bangladesh: A Look at Smart Grid Technology

B M Sarzil Redwan
Dept. of BECM, RUET



Bangladesh is one of the most populated countries and the most competitive industrial sector in South Asia after India.[1] By 2030, the country's economic growth of over 7% will require an estimated 34,000 MW of power.[2] The power generation capacity at present is 26,844 megawatts.[3] However, It is concerning that in Bangladesh, over 60% of power production comes from natural gas, while renewable sources contribute less than 2%.[4] According to its Intended Nationally Determined Contribution (INDC), the country plans to increase its share of renewable energy to 17% by 2041. This commitment aims to reduce greenhouse gas emissions by 5% until 2030.[5] Moreover, Bangladesh has another problem. Bangladesh's system loss is still high—at present, it is 7.92%.[6] Moreover, Power demand is increasing by 100MW per day, and the country may experience more than 3000 MW of load shedding this year. [7] This phenomenon is the main concern of Bangladesh's power sector. Smart grid technology can be a great solution to overcome these.



Fig: Smart Grid System

A smart grid is an advanced electricity network that utilizes digital technology for two-way consumer communication. This system enhances efficiency, reduces energy consumption, and improves transparency and reliability. To understand the smart power grid, let's first explore the conventional power grid. Its primary function is to transmit electricity from power plants to load centers. Transformers elevate voltage to minimize losses during long-distance transmission. High-voltage power travels through transmission lines, while additional transformers reduce it for consumer distribution. Interconnected power plants and load centers enhance grid resilience. Power utilities manage and operate these grids. A smart grid is an improved version of the existing power grid.

The first step towards improvement is the addition of a Fiber Optic cable. The Fiber Optic cable is placed on power transmission towers alongside power line conductors. This advanced communication channel opens up immense possibilities for electricity energy management. Additionally, the smart grid encourages the growth of small, medium-scale, and household power generation plants, such as solar and wind power plants. These distributed plants reduce dependency on major power stations and are often closer to consumers. Adding small, local power sources in the smart grid reduces transmission distances, lowering costs and losses. These distributed sources can fulfill electricity demand even if major power plants fail, minimizing consumer power cuts. Eventually, sufficient local power plants may independently meet local demand.

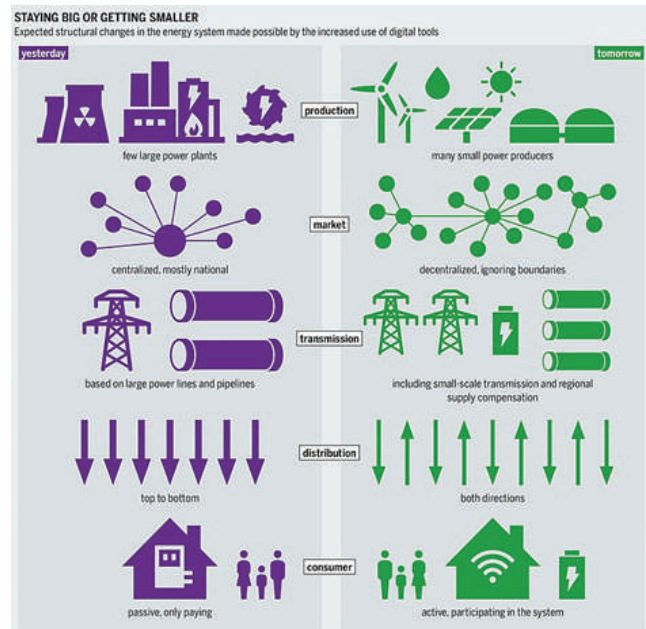


Fig: Characteristics of a traditional system

Developed nations in North America, Europe, and China actively implement smart grid technologies. Pioneering projects like Italy's Telegestore (2005) was completed in 2005, saving 500 million euros at a cost of 2.1 billion.[8] Austin, Texas, smart meter rollout (2003) demonstrates significant progress in the USA. These deployments manage millions of devices, enabling real-time monitoring and improved energy efficiency. Additionally, large-scale initiatives in Canada and Germany showcase the potential of smart grids for broader customer bases. These advancements contribute to environmental goals by facilitating bi-directional power flow, renewable energy sources, and intelligent energy management systems. However, as a developing nation, Bangladesh is lagging behind in smart grid technologies.

In the context of Bangladesh, smart grid technology will benefit economically. Smart grid technology can reduce economic loss and investment. The present electric grid was built in the 1900s and went through improvements as technology advanced through each decade. Smart grid technology is user-friendly as it enables consumers to create their own Home Area Network (HAN), which offers various practical applications. However, security is a major concern in this system, particularly in homes and offices. In case of any security issue, safety alerts will inform people who are away from home at that time. Additionally, consumers can reduce their electricity costs significantly by reducing peak loads in smart buildings. According to nationwide pilot data, consumers can reduce their electricity consumption by up to 25% during peak hours. Smart buildings, homes, and offices can also encourage consumer energy savings. The Bangladesh government can implement smart buildings with smart grid technology to minimize economic or financial loss.

Development is an ongoing process, and for Bangladesh, developing a smart grid should be considered an opportunity to improve the power sector. However, there are several challenges to implementing a smart grid in Bangladesh.

Firstly, policymakers are not aware of the benefits and necessity of a smart grid, so engineers must educate them. Secondly, financing the development of a smart grid will require substantial funding, so an initial pilot project with Bangladeshi and foreign investors could be undertaken. If successful, the government can consider expanding the smart grid. Thirdly, corruption is a concern when implementing large-scale projects, so policymakers should establish an independent evaluation committee free from external influence. The committee should comprise honest national and international bureaucrats, relevant ministry representatives, and Members of Parliament. Fourthly, policies and regulations for smart grid implementation need to be established. The Bangladesh Energy Regulatory Commission (BERS) should proactively revise existing rules and create new policies. They can draw inspiration from the Energy Independence and Security Act of 2007 (EISA), which addresses the modernization of the U.S. electric utility transmission and distribution system through smart grid upgrades. Lastly, training programs and workshops should raise consumer awareness about smart grids. These can educate consumers about smart grid usage, benefits, and potential implementation challenges.

The exciting news is that Bangladesh will soon enter the smart grid system, which is still in the initial stages. The 'Power Factor Improvement and Smart Grid under Dhaka Power Development Company (DPDC)' represents a blended finance initiative, combining an EU grant with a soft loan from the French development agency, AFD, totalling €112 million.[9] As part of this project, seminars are being organized for university and school students. These seminars facilitate learning, skill acquisition, knowledge retention, and practical application. By engaging students



Fig: Teesta Solar Limited, the country's largest solar power plant

in this revolutionary journey, the project aims to enhance the country's power system and promote smarter energy management. Furthermore, Bangladesh has made significant strides in increasing renewable energy capacity over the past few years, both at the national and consumer levels. This upward trend is likely to foster the adoption of smart grid systems.

Smart grid technology offers a compelling solution to Bangladesh's growing power demands and inefficiencies. While challenges exist, including policy awareness, financing, and consumer education, successful pilot projects and international collaboration demonstrate their viability. Embracing smart grid technology gives Bangladesh a unique opportunity to modernize its power sector, enhance energy security, and promote economic growth while transitioning towards renewable energy sources.



Did You Know??

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THE GREAT PYRAMID OF GIZA WAS THE WORLD'S TALLEST STRUCTURE FOR OVER 3,800 YEARS, BUILT FROM OVER 2,300,000 LIMESTONE BLOCKS.

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THERE ARE MORE THAN 180 MILLION PEOPLE WORKING WORLDWIDE IN CONSTRUCTION INDUSTRY, WHICH IS GREATER THAN THE ENTIRE POPULATION OF BANGLADESH!

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Tokyo is the most expensive city in the world to build, with an average cost of \$4,002 per sqm closely followed by Hong Kong at \$3,894 per sqm and San Francisco at \$3,720 per sqm



The Double-Edged Sword of Construction Mismanagement: Financial Strain and Environmental Damage

Abdur Rahaman
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Construction projects, in wide sense, are complex endeavors requiring meticulous planning, coordination, and execution. Mismanagement in construction can lead to a multitude of problems, including cost overruns, schedule delays, worker disputes, and improper resource allocation. Additionally, such mismanagement has significant negative impacts on the environment, causing significant environmental harm including air pollution and material waste.

Impact on Project Outcomes:

Cost Overruns: Mismanagement often leads to substantial cost overruns. Poor budgeting, unexpected changes in scope, and inefficient use of resources can inflate costs. For instance, failing to accurately estimate the materials required can result in either shortages, causing delays and additional costs to procure more, or surpluses, leading to wasted expenditure.

Schedule Delays: Projects frequently fall behind schedule due to inadequate planning, poor coordination among subcontractors, and unforeseen issues that could have been mitigated with better foresight. Delays not only increase labor costs but also extend the period during which the project's financing accrues interest, further escalating costs.

Improper Resource Allocation: Mismanagement can result in improper allocation of resources, where materials, labor, and equipment are not optimally utilized. Improper waste management leads to overflowing landfills and potential contamination of soil and water resources.

Safety Hazards: Rushed projects or inadequately trained workers due to mismanagement can lead to safety hazards. Cutting corners on safety protocols to meet deadlines significantly increases the risk of accidents and injuries, jeopardizing worker well-being and potentially incurring legal ramifications.

Project Dysfunction:

Communication Breakdown: A major culprit in mismanagement is poor communication. Unclear instructions, lack of collaboration between stakeholders, and information silos can lead to misunderstandings and mistakes. This can result in wasted resources, rework, and frustration among team members.



fig: Improper management of construction materials.

Low Morale: When projects are plagued by delays, cost overruns, and safety concerns, worker morale suffers. Disorganized work environments and a lack of clear direction can lead to disengagement and a drop in productivity. The ultimate victim of mismanagement is often the client. Budgetary constraints, delays in completion, and substandard work can lead to significant dissatisfaction and damage the reputation of the project team. In worst-case scenarios, legal disputes can arise, further adding to the project's woes.

Environmental Impact:

Air Pollution: Construction activities generate significant dust and emissions from machinery, contributing to air pollution. Poorly managed projects can exacerbate this issue through inefficient machinery use and extended project durations, resulting in prolonged periods of pollution. Additionally, the increased need for corrective work often means more dust and emissions as work is redone or adjusted.

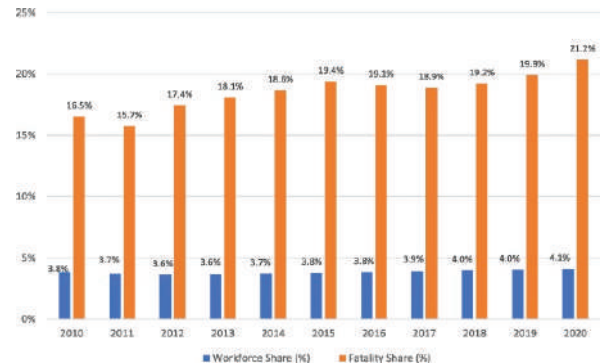


fig: Construction Workforce and Fatality Shares due to mismanagement (2010-20)

Waste Generation: Ineffective planning and resource management often lead to excess material wastage. Materials like concrete, steel, and timber, when over-ordered or improperly stored, contribute to landfill volumes and resource depletion. This wastage not only represents financial loss but also has a detrimental impact on the environment.

Soil and Water Pollution: Construction sites are prone to generating pollutants such as oils, chemicals, and hazardous substances. Mismanagement in handling these materials can lead to spills and leaks, contaminating nearby soil and water bodies. Poorly managed stormwater run-off can carry these pollutants into rivers and lakes, harming aquatic ecosystems.

Conclusion: Mismanagement in construction projects presents a dual threat, significantly impacting both the project's success and the environment. Financial losses due to cost overruns and schedule delays, combined with worker disputes and safety issues, it undermines the integrity of construction initiatives. To mitigate these issues, it is imperative for construction managers to adopt comprehensive planning, effective communication, and stringent environmental safeguards. By addressing these areas, the construction industry can not only enhance project outcomes but also contribute to a more sustainable and resilient environment.



Towards Sustainable Construction: Environmental Challenges in Bangladesh

Sultana Musfirat Negar Parvin

Dept. of BECM, RUET

The unplanned growth and high population density of Asian cities lead to environmental damage. The high population density, poverty, and pollution in Bangladesh make it more problematic. In order to solve the issue of environmental deterioration, government initiatives centered on commissions and worldwide studies of the impact of development are critical. Due to the delicate balance that exists between humans and their environment, addressing recurring environmental catastrophes demands a comprehensive plan. Since sustainability is today a major concern and calls for proactive measures to minimize environmental effects and ensure project success, building and development are closely intertwined.

Examining environmental problems that Bangladesh's construction sector faces is the aim of this study. The specific objectives include analyzing the environmental effects of the sector as a whole, examining the rules and regulations that are in place at the moment, assessing the impact of climate change on built environments, and explaining the reasoning behind environmental action in the construction industry. This study attempts to clarify how environmental concerns could be handled in the construction sector, with particular emphasis on Bangladesh.

The building sector uses over 50% of all non-renewable resources worldwide, which makes it incredibly unsustainable. Despite the ecological costs, humanity has always tried to alter the environment to fit its wants. This has resulted in a way of life that is deeply integrated throughout many different kinds of buildings. Although buildings are essential to modern society, the rate at which resources are being used is unsustainable for the planet.

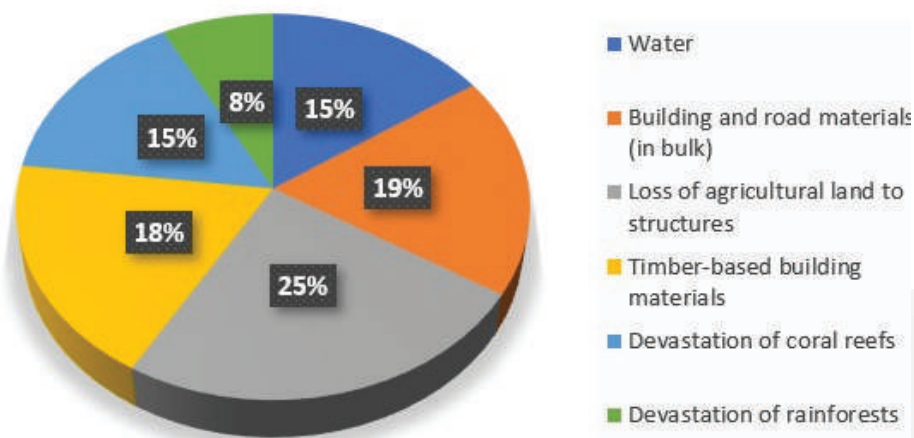


fig: An estimate of the resources used in construction (Rasel & Amin, 2021)

Living and working buildings are constructed in response to the demands of urbanization and industrialization, which have an impact on the environment, especially during the building process. Gaining knowledge of these implications is necessary for effective mitigation. Degradation of habitat, resource depletion, and pollution are significant issues that require waste management and pollution control. Sustainable building methods take into account biodiversity and climate change while maintaining the integrity of natural systems for a sustainable future. Adherence to environmental standards and stakeholder cooperation are essential.

To reduce environmental difficulties in Bangladesh's construction sector, the following solutions can be implemented:

Alternative Building Materials: Promoting the use of non-burnt bricks and concrete blocks to reduce topsoil extraction and air pollution.

Regulatory Measures: Enforcing laws to prevent soil extraction from agricultural lands and encouraging sustainable practices.

Sustainable Practices: Using recycled materials, improving energy efficiency, and adopting green building certifications to reduce environmental impact.

Technological Innovations: Investing in prefabricated construction and new technologies to minimize waste and enhance efficiency.

Awareness and Training: Educating stakeholders on sustainable construction benefits and providing training on environmental regulations.

International Support: Collaborating with international organizations for technical and financial assistance in sustainable construction projects.

The production of waste and air pollution are two important environmental repercussions of building. Mitigation strategies like Environmental Impact Assessments (EIAs) and Sustainable Construction are hard to enforce. It is vital to find long-term supervision, legislative support, harsh penalties, open contracts, and career growth. As soon as the project commences, it is imperative to include these standards-related measures. Rules must be strictly followed and their negative impacts on the environment must generally be reduced.



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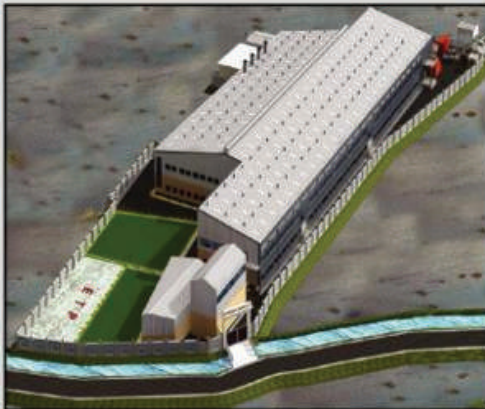
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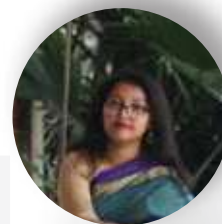
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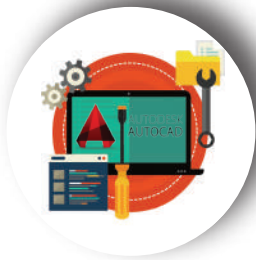
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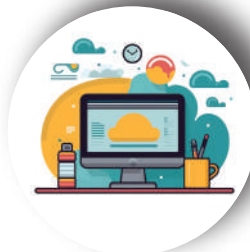
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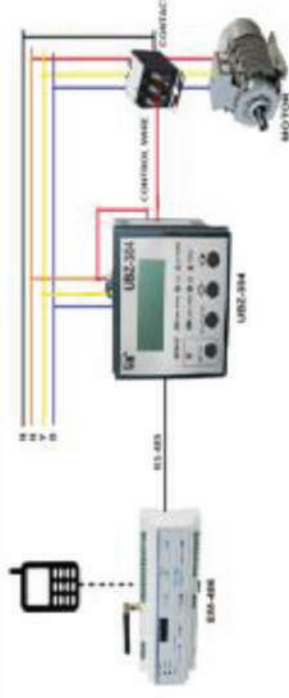
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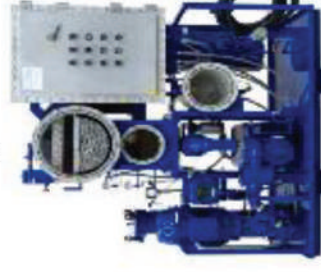
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